Recitation 4

Relations and Functions

Part 1: Relations

Definitions

Defn 1: A *relation* $R : A \to B$ is defined on a domain A, co-domain B, and a graph that is a subset of the Cartesian product $A \times B$. As a slight abuse of notation, we will often write the relation R as a subset of $A \times B$. A relation on a set A is $R : A \to A$.

Defn 2: An *equivalence relation* is a relation that is reflexive, symmetric, and transitive.

Defn 4: A relation R on A is reflexive if $\forall a \in A$, $(a, a) \in R$.

Defn 5: A relation R on A is *symmetric* if $\forall (a, b) \in R$, $(b, a) \in R$. An equivalent definition is that a relation is *not* symmetric if $\exists (a, b) \in R$ such that $(b, a) \notin R$.

Defn 6: A relation R on A is *antisymmetric* if $\forall a, b \in A$, $(a, b) \in R$ and $(b, a) \in R$ implies that $a = b$.

Defn 7: A relation R on A is *transitive* if $\forall (a, b), (b, c) \in R$, $(a, c) \in R$. An equivalent definition is that a relation is *not* transitive if $\exists (a, b), (b, c) \in R$ such that $(a, c) \notin R$.

Defn 8: Let R be an equivalence relation on A. Then, the *equivalence class* of $a \in A$, denoted 4 $[a]_R$, is $\{x \in A \mid (a, x) \in R\}$. That is, $[a]_R$ is all of the elements to which a is related.

What is the point of an equivalence relation, anyway?

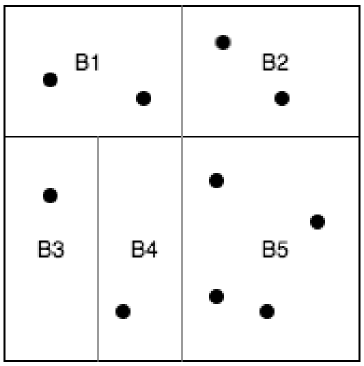
What does it mean for two things to be equal? It can depend on context. For example, you probably generally think of the numbers 2 and 4 as not being equal. However, maybe I want to consider the numbers 2 and 4 to be equal in some contexts because they are both even. We could be in a situation where we only want there to be two kinds of things: even things and odd things. We don't care about anything else like how big or how small the thing is.

An equivalence relation allows us to specify what things in the world are equal to each other, and what things aren't.

An equivalence relation R splits up our world into categories, or equivalence classes. In a given equivalence class, all things within the class are things we consider equal, or equivalent, in the context of R.

For instance, the equivalence relation $R = \{(x, y) \mid x \bmod 2 = y \bmod 2\}$, for all $x, y \in \mathbb{Z}$, divides the world into two categories: the odd and even numbers.

We call the way the equivalence relation splits up our world a *partition*. A little more formally, a *partition* of a set A is a list of subsets $B_1, \ldots, B_k$ of A such that every element of A is in some subset B_i (exhaustive), but no two subsets share an element (mutually exclusive).



One possible partition of some set A, where the dots in the square are distinct elements of A

Task 1

a. Consider the set $A = \{1, 2, 3\}$. In the following questions, all relations are on A. It may be helpful to draw out a diagram of each relation.

   i. $R = A \times A$. List out the elements of R. Is R an equivalence relation? If so, state its equivalence class(es).

   ii. $R = \{(1, 2), (2, 1)\}$. Is this relation *transitive*?

   iii. $R = \{(1, 2), (2, 1), (2, 2), (1, 1)\}$. Is this relation reflexive? Symmetric? Transitive?

iv. If the relation in question iii is not an equivalence relation, can you add one pair to it and make it an equivalence relation? Write the equivalence classes of the new relation.

b. Let $A = \{1, 2\}$ and answer to the following questions.

(a) What is the equivalence relation on A with the smallest number of equivalence classes possible?

(b) What is the equivalence relation on A with the largest number of equivalence classes possible?

(c) Is $R_0 = \{\}$ a relation on A?

(d) Is R_0 symmetric? Is it antisymmetric? Why or why not?

(e) Is R_0 transitive? Why or why not?

(f) R_0 is not an equivalence relation. Why?

c. Suppose R is an equivalence relation on S, and $R = \{\}$. What is S?

d. Consider the set B of all students at Brown. For each of the following relations on B, state whether they are reflexive, symmetric, antisymmetric, transitive, or some combination of them. If it is an equivalence relation, then determine the equivalence classes of the relation.

i. Two students are related if they have the same astrology sign.

ii. s_1 and s_2 are students and $(s_1, s_2) \in R$ if s_1 is younger than or the exact same age as s_2. (You can assume no students were born at the exact same time.)

iii. Two students are related if they are studying anthropology.

iv. Two students are related if they go to Brown.

Checkpoint 1 — Call a TA over!

Part 2: Functions

Definitions

Defn 1: A relation $R : X \to Y$ is a **function** if for every x in the domain X, x is mapped to one and only one y in Y, the codomain. Note that in the book this is called a *total function*, and function refers to a *partial function*, where for every x in the domain X, x is mapped to zero or one y in the codomain Y. In this class, we will use function to mean total function and partial function to mean partial function.

Defn 2: The **range** of a function f consists of all members of the codomain of f that are mapped to by some member of the domain of f. It is the *image* of the domain.

Defn 3: $f : X \to Y$ is **injective (one-to-one)** if, for every $y \in Y$, there is *at most one* $x \in X$ such that $f(x) = y$. Equivalently, for any $x, y \in Y$ we have $f(x) = f(y) \implies x = y$, and you can also use its contrapositive $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$.

Defn 4: $f : X \to Y$ is **surjective (onto)** if, for every $y \in Y$, there is at least one $x \in X$ such that $f(x) = y$. For surjective functions, the range is equal to co-domain.

Defn 5: $f : X \to Y$ is a **bijection** if it is both an injection and surjection.

Task 2

Let A be the set $\{1, 2, 3\}$. Consider the following relation on A, $R_1 = \{(1, 2), (2, 1)\}$.

1. Is R_1 a function?

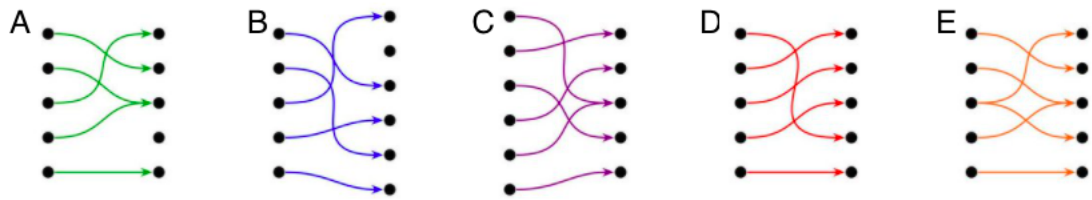
Now, consider R_2, another relation on A: $\{(1, 2), (2, 1), (3, 2)\}$.

1. Is R_2 a function?

2. If R_2 is a function, what is its codomain? How about its range?

Task 3

Consider these diagrams that visualize a relation $R : A \to B$. The diagrams have two sets of dots, one for A and one for B, and they have an arrow from a to b in whenever $(a, b) \in R$.



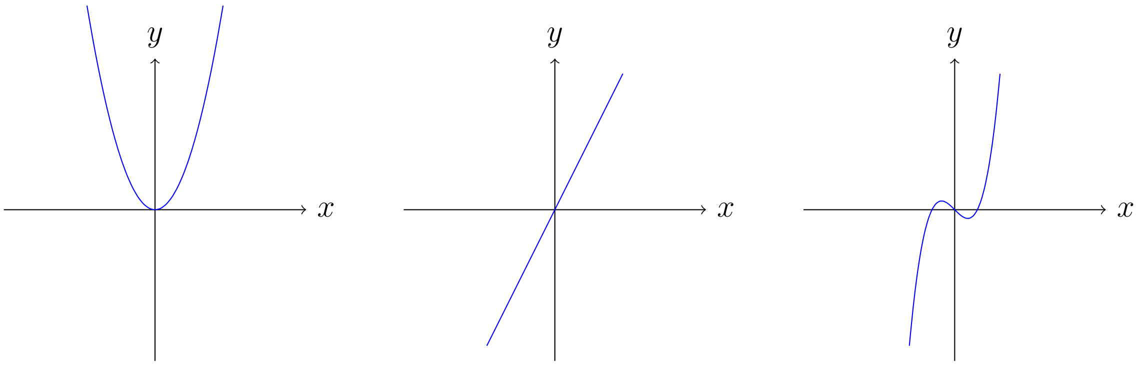
Match each of the five diagrams, labeled A–E, with one of these five descriptions below:

1. ___ Not a function
2. ___ A function that is neither surjective nor injective
3. ___ A surjective function that is not injective
4. ___ An injective function that is not surjective
5. ___ A bijective function — both surjective and injective

Optional Checkpoint (recommended if queue is short) — Call a TA over!

Task 4

Consider the following functions and determine if the given function is an injection, surjection, and/or bijection.



(a) $f(x) = x^2$ (b) $g(x) = x/2$ (c) $h(x) = x^3 - x$

Discuss your answers!

a. $f : \mathbb{R} \to \mathbb{R},\ f(x) = x^2$

b. $g : \mathbb{R} \to \mathbb{R},\ g(x) = \frac{x}{2}$

c. $h : \mathbb{R} \to \mathbb{R},\ h(x) = x^3 - x$

d. Question: All of the above functions are defined on $\mathbb{R}$. Consider their graphs in the coordinate system. Which of the following implies surjectivity, which implies injectivity, and which implies neither?

(1) any vertical line intersects the graph **at most** once

(2) any horizontal line intersects the graph **at most** once

(3) any vertical line intersects the graph **at least** once

(4) any horizontal line intersects the graph **at least** once

e. Recall the functions defined in parts a-c. Is $f \circ h : \mathbb{R} \to \mathbb{R}$ surjective and/or injective? (Use a graphing calculator if you need to.)

f. Is $h \circ f : \mathbb{R} \to \mathbb{R}$ surjective and/or injective?

g. Let $f : \mathbb{R} \to \mathbb{Z}$ give as output the greatest integer less than or equal to x, denoted as the floor function $f(x) = \lfloor x \rfloor$. For instance, $f(3.5) = 3$, $f(3) = 3$, and $f(\pi) = 3$.[1]

h. $f : \mathbb{Z} \to \mathbb{Z}$

$f(x) = \lfloor x \rfloor$

i. *Optional:* f : Brown University Students $\to$ Countries in the World

f(student) = country where student is from

j. *Optional:* f : First Year Students $\to$ First Year Dorms

f(student) = dorm that student lives in

[1] Note that we can similarly define the ceiling function $f : \mathbb{R} \to \mathbb{Z}$ that gives as output the smallest integer greater than or equal to x, denoted the ceiling function $f(x) = \lceil x \rceil$. For example, $hf(3.5) = 4$, $f(3) = 3$, and $f(\pi) = 4$.

Task 5

a. Let $A = \{0, 1, 2\}$ and the function $f : \mathcal{P}(A) \to \{0, 1\}^3$.

Denote the output as (f_0, f_1, f_2), where each f_i is 0 or 1. Given an element $X \in \mathcal{P}(A)$, define $f(X)$ where each f_i is 1 iff $i \in X$, and 0 otherwise.

For instance, $f(\{0, 2\}) = (1, 0, 1)$, $f(A) = (1, 1, 1)$, and $f(\emptyset) = (0, 0, 0)$.

i. Consider a set of size n where $A = \{1, 2, \ldots, n\}$. Generalize the above bijective function for $\mathcal{P}(A)$ and binary strings of length n (that is, $\{0, 1\}^n$).

ii. Prove that the above function is a bijection.

b. If $f : X \to Y$ is bijective, and $g : Y \to Z$ is bijective, what can we say about the cardinalities of X and Z? Can you say that $g \circ f$ is bijective?

Optional Task 6

a. If a function $f : X \to Y$ is injective, what can we say about the cardinalities of X and Y? Try making some diagrams where X has more elements than Y, fewer elements than Y, or the same number of elements as Y. When are you able to create an injection, and when are you not?

b. If a function $f : X \to Y$ is surjective, what can we say about the cardinalities of X and Y? Again, you might want to draw out some examples.

c. Based on what you've found in the previous two questions, if a function $f : X \to Y$ is bijective, what can we say about the cardinalities of X and Y? When can we create a bijection between two sets, and when can we not?

Checkoff - Call a TA over!